International Baccalaureate Extended Essay

Investigation on the optical power loss of fiber optics.

Subject: Physics

Research question:

To what extend does lateral displacement affect the power received by a single-mode fiber when aligned with another single mode fiber emitting a beam of 4 mW.

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Contents

Introduction 3

Theory 4

1.1 Axis and terms 4

1.2 Gaussian beam intensity distribution 5

1.3 Propagating beam radius 6

1.4 Gaussian beam function: 7

1.5 Fresnel reflection 7

1.6 Mode matching (overlap integral) 8

Methodology 10

2.1 Experiment setup 10

2.2 Independent variables: 11

2.3 Dependent variables 12

2.4 Controlled variables: 12

2.5 Apparatus: 12

Data collection 13

3.1 Method of data collection: 13

3.2 Raw data 14

3.3 Analyzed data 15

Analysis 15

Conclusion and Evaluation 18

6.1 Weaknesses: 18

6.2 Strengths: 19

6.3 Further improvement/investigation: 19

References 20

Appendix: 21

Introduction

The fiber optic is a hair thin silica cable that transports coded light inside with the principle of total internal reflection. Since its introduction it has become arguably humanities most important method of transporting information with exponentially growing importance due to optic fibers characteristics of having a theoretically capability of “almost infinite” bandwidth due to the fact that multiple wavelength signals can travel and carry information simultaneously. With the rise of ai and the need for faster data processing centers the future is very exciting for fiber optics as this field could very well become one of the most sought-after technologies of the coming decades. With the first working fiber optic only being patented in 1966 (Ghatak) and thus the prospect of many more discoveries in this field to lie ahead instead of behind of us it has sparked an interest to learn about it. Today almost all cables providing internet are fiber optics (Griffiths) they were determined to be the most suitable because aside from its fast speed and bandwidth compared to other media, light does not interact with electromagnetic fields nor does it react to moisture and temperature making it ideal for long range underwater communication running the internet.
The two most common types of fiber are single-mode and multi-mode fibers, both with its distinct uses. The single-mode fiber has a core diameter of around 9 micrometers which can transmit around 10 gigabits per second per wire and is designed as a long-range low attenuation medium which ideally only has to one wavelength or mode this minimizes modal dispersion and allows a single signal to be able to transmit a few hundred kilometers before needing an amplifier. These are buried below the ocean for long range communication. The second type are multi-mode fibers they have a wider core of around 60 micrometers and can transmit great amounts of modes within a wavelength range that the technology allows it to have which allows some types to have a bandwidth of around 100 gigabits per second per

fiber. These are the fibers which are commonly used inside data bases and similar facilities.

While both are fascinating, I intend to only discuss the single-mode variant as the two types do quite differ in many ways. The single-mode fiber was chosen due to its relative simplicity in emission pattern as well as a smaller core diameter so that any measurement ranges can be smaller compared to multi-mode fibers. This will be my introduction to fiber optics so the goal of this extended essay will be to learn about fiber optics and more specifically about how the laser beam propagates once it leaves the fiber. As well as answer my research question of how during fiber coupling, or the aligning of fibers and then gluing them together how the misalignment would affect the optical power. Or in other words:

To what extend does lateral displacement affect the power received by a single-mode fiber when aligned with another single mode fiber emitting a beam of 4 mW.

Theory

1.1 Axis and terms

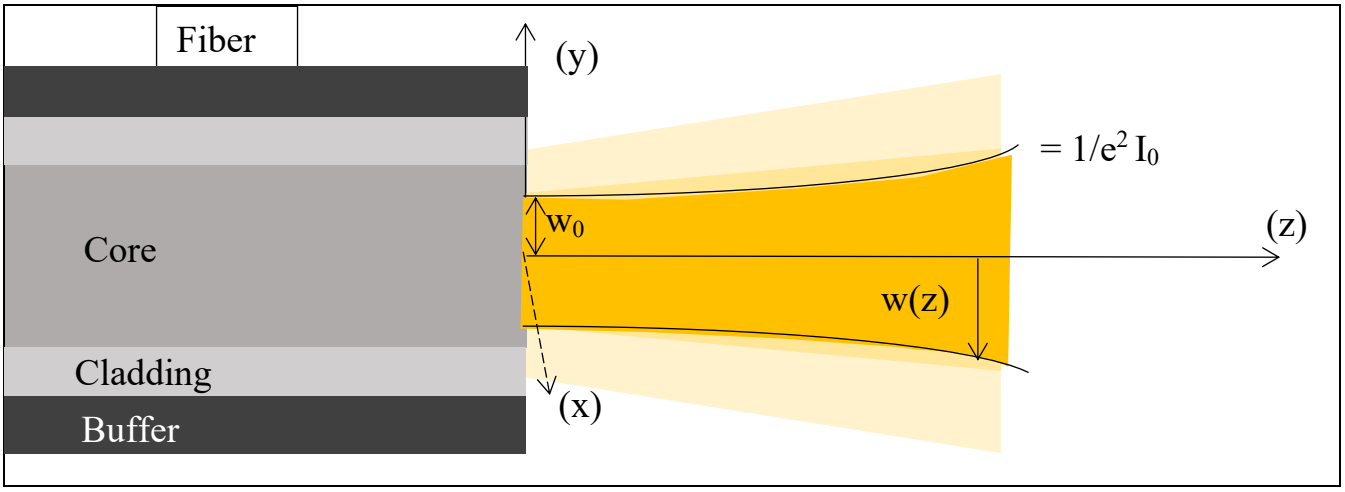


Figure 1: diagram of beam leaving fiber. (self-drawn)

- Lateral displacement axis (z): is labeled as the direction in which the wave propagates with x and y being the orthogonal directions.
- I_0: is the intensity at the origin (0,0,0).
- The radius (r) is the distance between the center and a point in the same x-y plane
- Waist radius/Mode field diameter (ω_0): Is the distance (r) from center axis to where the wave intensity is $1/e^2\, I_0$ when the beam exits the fiber. Which is 9.2μm for a Corning fiber (Vastag).
- Radius (ω): is the distance (r) between the center axis (z) and where the wave intensity is $1/e^2\, I_0$ in that plane.
- Initial optical power (P_0): is the amount of power supplied which is 4mW

1.2 Gaussian beam intensity distribution

As the beam I am studying is propagating in a cylindrical homogenous cavity in other words the beam is propagating through a continuous medium which has a circular surface. Since fibers utilize the phenomena of total internal reflection negligible light exit the cladding due to being reflected. As fibers are always bent even a perfect power source emitting a perfectly straight laser will get scattered as some light reflects of the cladding and changes directions, still forwards but at a different angle. As there are large amounts of photons traveling inside the fiber, we can use the probability distribution of the random movements of the photons as the intensity distribution since the intensity is proportional to the number of light rays reaching that point. As a normal distribution is in the form of

$$P(x) = a\exp(\frac{-(x-b)^2}{2c^2})$$

Where a, b and c are constants depending on what conditions there are for the normal distribution.

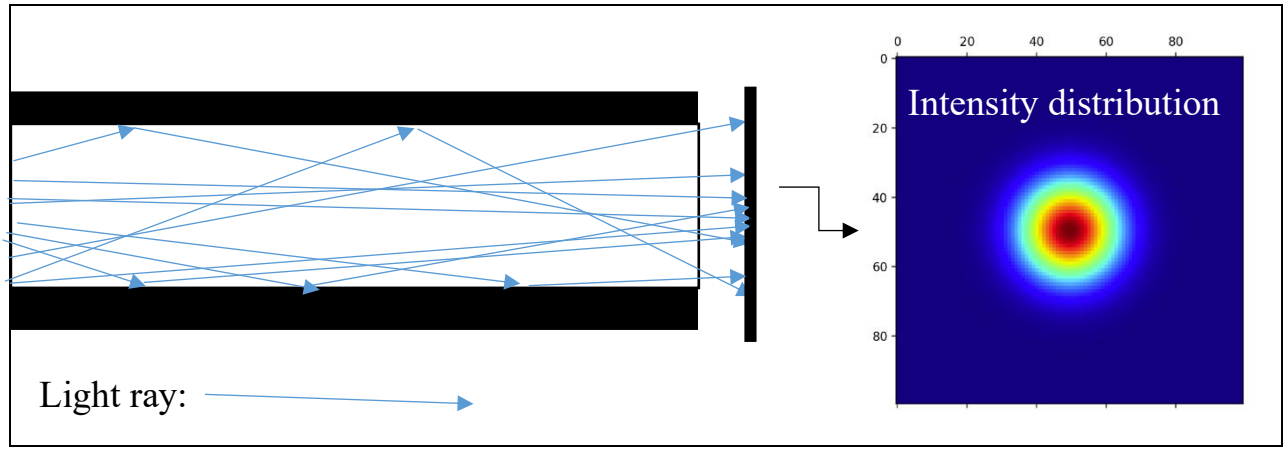


Figure 2: the scattering of light rays when it is in the fiber and the following intensity distribution caused by this distribution. (self-drawn)

The normal distribution（Gaussian distribution）holds for almost all laser beams (Paschotta), beams that follow this intensity distribution caused by the scattering of light are called Gaussian beams. A beam that comes out a fiber optic has the intensity distribution:

$$I(r,z) = \frac{2P_0}{\pi\omega(z)^2} \exp\left(\frac{-2r^2}{\omega(z)^2}\right) \quad (1)$$

(Natividad et al.)

1.3 Propagating beam radius

Due to diffraction no beam will remain in the same shape when propagating. The beam radius of any beam will change when propagating. This can be described mathematically as.

$$\omega(z) = \omega_0\sqrt{1+\left(\frac{z\lambda}{\pi{\omega_0}^2 n}\right)^2} \quad (2)$$

(Svelto)

Where z is the distance the beam has propagated and $\omega(z)$ is the radius of the beam at a given z.

1.4 Gaussian beam function:

Combining (1) and (2) we get:

$$I(r,z) = \frac{2P_0}{\pi\omega_0{}^2\left(1+\left(\frac{z\lambda}{\pi\omega_0{}^2 n}\right)^2\right)} \exp\left(\frac{-2r^2}{\omega_0{}^2\left(1+\left(\frac{z\lambda}{\pi\omega_0{}^2 n}\right)^2\right)}\right) \quad (3)$$

This is the intensity distribution of a gaussian beam depending on its distance of propagation (z) and its distance to the center of the beam (r).

1.5 Fresnel reflection

A point of interest is that the power measured when the source and receptor fiber are "aligned" is significantly lower than the power received at 10μm displacement. Which would be violating the law of conservation of energy as the energy of the beam could not increase during propagation without another source of energy.

Due to this irregularity, I also measured the exact power of the source fiber by directly inserting the source fiber into the power receiver measuring 6.01 dBm or 4 mW. The difference is because of power losses during the transmission from source to receiver. The two main reasons are: Is misalignment and Fresnel reflection, other causes are attenuation, cleanliness of the fiber head fiber bend, imperfect materials. When light exits the fiber, it is partially reflected by the transition of mediums by Fresnel reflection, the percentage reflected (R) can be calculated by the following equation (Paschotta):

As a beam propagates from a medium into a less dense medium some partial reflection will occur. As every light wave experiences this medium change the reflected amount will be a percentage of the overall beam energy. The percentage will only depend on the two media and the angle of incidence as shown in the equation below. With θ being the angle of incidence or refraction (Paschotta).

$$R = \left(\frac{n_2\cos\theta - n_1\cos\theta}{n_2\cos\theta + n_1\cos\theta}\right)^2 \quad (4)$$

With R being the percentage of light reflected

With the refractive index of fiber being 1.4674 and the angle θ being 0:

$$\mathrm{R} = \left(\frac{1.4674 - 1}{1.4674 + 1}\right)^2 = 0.0359$$

4% get reflected during each transition, once when exiting the source fiber to air and once entering the receiver fiber from the air. The maximum power that could be received even with a gap can found:

$$P_{eff} = P_0(1 - R)^2 = 3.718 \text{ mW}$$

P_{eff} is the effective power which can be absorbed if all non-reflected light gets absorbed by the receiving fiber.

1.6 Mode matching (overlap integral)

The theoretical power the fiber should receive can be derived theoretically. The emission distribution of a fiber optic is the same as the absorption distribution of a fiber optic. This makes sense as the path of light forwards and backwards is the same according to Snell's law. The power the receptor fiber can receive can be calculated by integrating the intensity distribution of the propagating beam at that z-position over the surface of the second fiber. The intensity distribution of the beam at the surface when it meets the receptor fiber is the Gaussian beam intensity distribution after it has partially reflected and propagated until the second fiber. And the absorption distribution at that surface of the receptor fiber is the gaussian intensity distribution of the fiber which is equivalent to the intensity distribution of the beam when it exits a fiber, which is:

$$\mathrm{I}(\mathrm{r}, 0)$$

The power that theoretically should get received is the overlap integral of the two intensity distributions multiplied with each other. The power received is the original power (P_0) multiplied with the overlap integral which is the following equation: (Savović et al.)

$$P = P_0 \left| \iint I(r,z) I(r,0)\, dA \right|^2 \tag{5}$$

I can calculate power received as $P = \int I\, dA$. As both fibers have a circular surface, I need to change the dA into polar coordinates. Using a Jacobian determinant r (Texas University), I can integrate the intensity of a line from the center to the radius of the fiber surface, where θ is constant. Then integrating all those lines for every θ or 2π radians. Since the line is itself irregular but each line of θ is the same for every angle because of the Gaussian beams property of being symmetrical so no term is dependent on θ. Thus, with adding boundaries to equation (5) and then transforming it into polar coordinates and adding r as the Jacobian determinant we can find the power a fiber of radius 4.1μm should receive if the beam is a gaussian distribution:

$$P = P_0 \int_0^{2\pi} \int_0^{4.1\mu m} \left| I(r,z) I(r,0)\, r\, dr\, d\theta \right|^2 \tag{6}$$

Since θ is unrelated:

$$P = 2\pi P_0 \int_0^{4.1\mu m} \left| I(r,z) I(r,0) r\, dr \right|^2$$

Substituting $I(r,z)$ with (3) we get:

$P =$

$$2\pi P_0 \int_0^{4.1\mu m} \left| \frac{2P_0}{\pi\omega_0{}^2\left(1+\left(\frac{z\lambda}{\pi\omega_0{}^2}\right)^2\right)} \; exp\left(\frac{-2r^2}{\omega_0{}^2\left(1+\left(\frac{z\lambda}{\pi\omega_0{}^2}\right)^2\right)} \right) \frac{2P_0}{\pi\omega_0{}^2} \; exp\left(\frac{-2r^2}{\omega_0{}^2}\right) r \; dr \right|^2 \qquad (7)$$

Methodology

2.1 Experiment setup

The plan is to collect data on the coupling efficiency of the two fibers when experiencing different amounts of misalignment. For this I will use a fiber coupling station that I got access to during a summer internship which for the sake of this experiment acts as a pair of very steady hands that can move the two fibers in all the cardinal directions with almost perfectly straight paths. By clamping the two fiber heads on the movable holder and turning the position dials it allows me to measure the effect of changing only one axis without displacing the fiber on another axis.

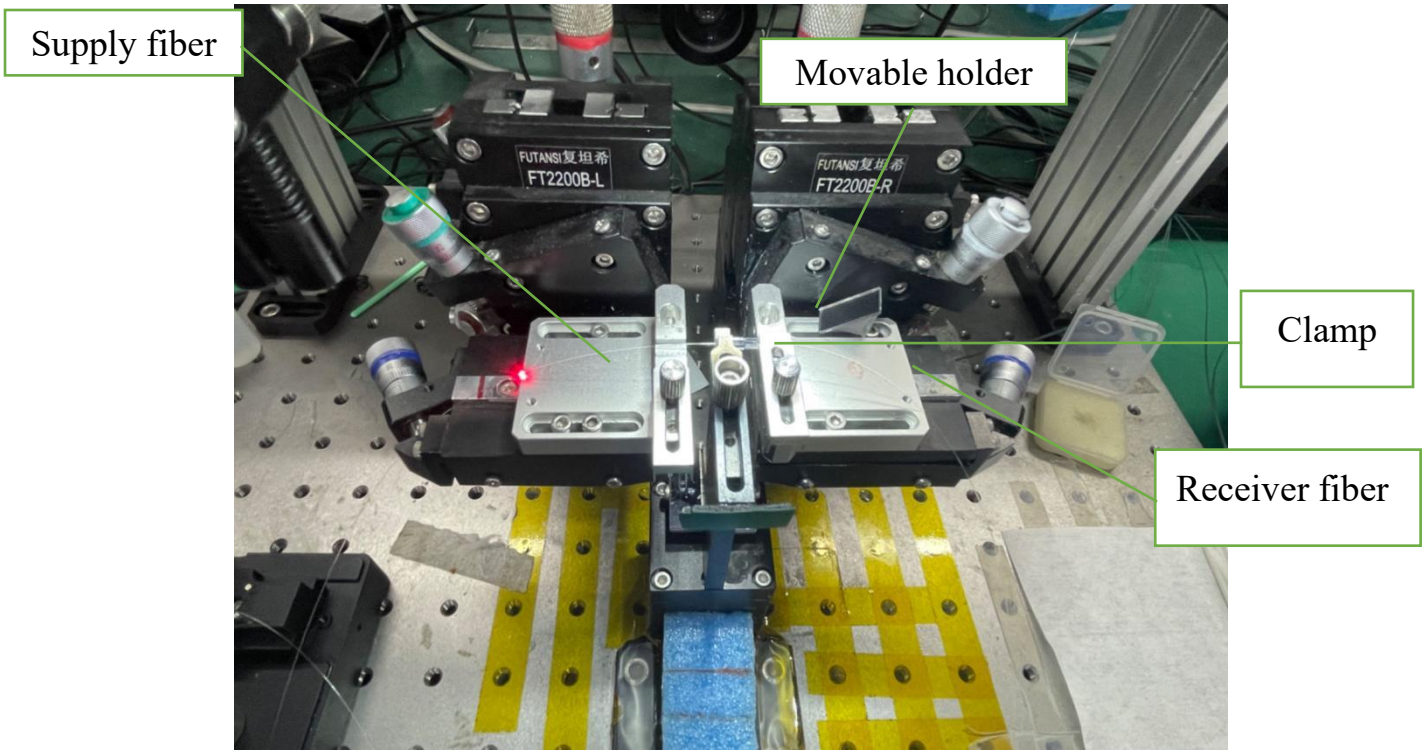


Figure 2: Apparatus Fiber aligning station. (Photographed by candidate)

There will be two factors that would affect the accuracy of the experiment that could be minimized with the appropriate method.

i. Firstly, the issue that all fibers have slightly different properties and surfaces, for this during each set the same fiber have to be used and when trying to measure the effects of misalignment I need to also measure the aligned data set with the same fiber so that a meaningful comparison can be made between the efficiency when aligned and when misaligned.
ii. Secondly the issue of initial alignment, since the scale is so small initial angular misalignment is impossible to spot on the power meter as well as on the camera system. I intend to solve this by initially moving a fiber away from the other by around 50 micrometers and utilizing the fact that the center of a beam has the highest intensity I will move that fiber around in that x-y plane until the power received is the largest it can be and then calculating the angle using $\theta = \tan^{-1}\frac{r}{z}$ and then adjusting by that using the angle dial.

Then I will compare my results to the theoretical results which will be interesting to see if how different ideal conditions are to real conditions and how the actual beam differs from the theoretical gaussian beam.

2.2 Independent variables:

I. The distance r or which is the distance to the central z axis in any plane.
II. The distance from the source or the z value. This is because by changing the z the waist radius of the beam will change and by measuring the power I could deduce if the change in power correlates with the theoretical waist radius. The range of the of the z-axis displacement will be from 0 to 100 micrometers as my pre-experiments have shown that after

2.3 Dependent variables

I. The power received will change based on the position of measurement

2.4 Controlled variables:

I. The optical power has to stay relatively constant.
II. The fiber properties have to stay constant, by using a corning smf-28 fiber the ω_0 can be calculated using the Gaussian beam radius formula.
III. Keeping the wavelength and medium of propagation constant so that the refractive index does not change.

2.5 Apparatus:

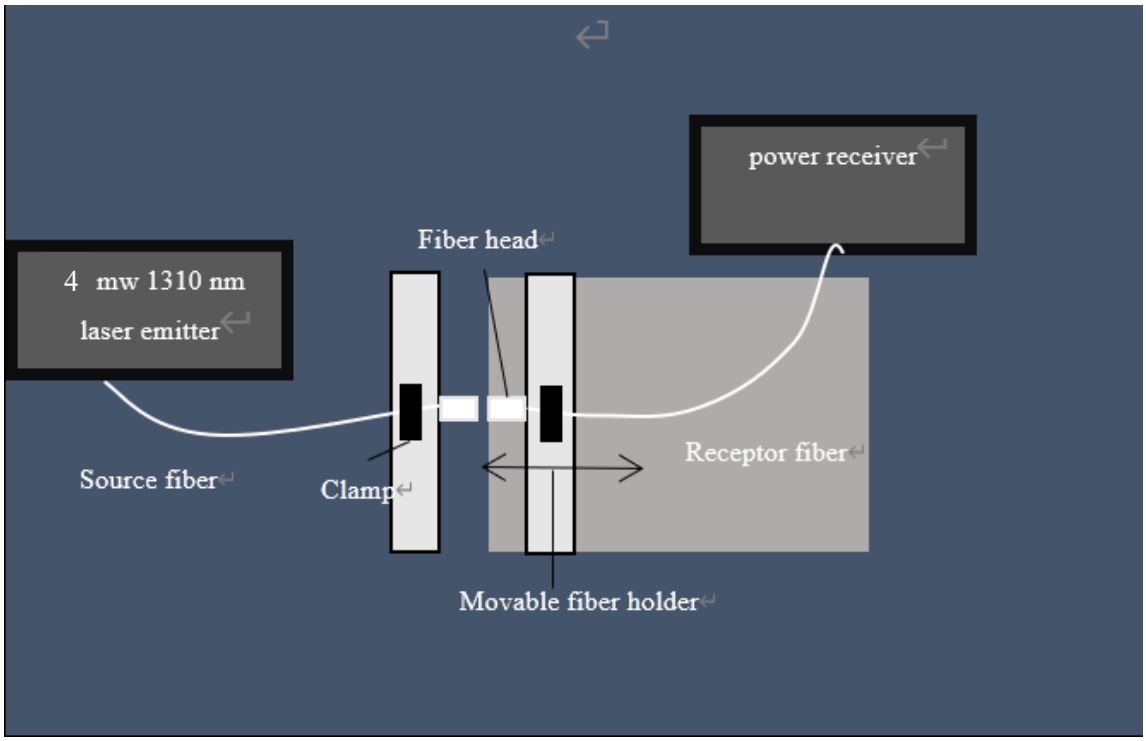


Figure 3: Coupling station top-down view. (self-drawn)

- laser emitter (monochromatic, 4 mW, 1310 nm wavelength)
- single mode fiber (8.2μm core diameter)
- movable fiber holder adjusted with rotational dial with graduates of 1 μm and range 1mm

- Power receiver with units in dBm ±0.01, but since the reading always fluctuates due to the changes in the power source so the actual uncertainty is ±0.05 dBm.

Data collection

3.1 Method of data collection:

The experiment consisted of a preparation phase and a measuring phase:

(preparation)

1. Before starting each set the fiber head has to be wiped using alcohol and polished.
2. Both fibers have to be placed corresponding to their slope so that they align. Note: an angled fiber head does not mean an angled fiber/beam, since the alignment in the next step will be determined the orientation of the fiber.
3. Aligning the fiber heads by turning each dial slightly and adjusting until finding the largest power value on the power receiver. Decreasing the initial misalignment. And recording the initial power received (P_0).
4. Then to find the actual axis of propagation the fiber has to be moved away and then changing the receptor fibers x and y axis until the power reading is at its maximum.

In order to find out the energy distribution a beam in 3d space I will need to first determine if the beam is symmetrical around its central z axis. My method will be by collecting the power output in two z-planes. If both planes are symmetrical the beam can be assumed as symmetrical in all planes in this magnitude of z-axis displacement.

The energy distribution of a plane can be measured as following:

5. Turn the z-axis dial by 4 rotations or 200 units.
6. Record the power value.
7. Turn the x-axis dial 5μm in the positive (fiber to the right) direction.
8. Wait until the power reading stabilizes.
9. Repeat step 6-7 until 5 times.
10. Repeat 6-8 but dialing in the negative direction (fiber to the left).
11. Return the x-axis dial to the origin.
12. Dial the y-axis by 5μm 5 times in both the positive (up) and negative (down) direction.
13. Repeat step 6-9 for every y-axis movement so a plane of data is collected.

Then to measure the energy received when the fibers have a gap displacement or a z-axis misalignment the following steps were taken:

14. Turn every dial to its starting position
15. Check if the power reading is the same as step 3. If less than ±0.5 dBm continue, if the more than 0.5 dBm then it means that the fiber moved with respect the clamp then restart.
16. move the z axis by 10μm and record the power at the z-axis
17. Collect the power values for the plane as described in step 13.
18. Record the location and the power at that coordinate.
19. Dial the x and y axis back to 0
20. Repeat for each z-axis displacement.

3.2 Raw data

Displacement Z/μm (±0.5μm)	trial	0.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
Power received P/dBm (±0.05dBm)	1	4.77	4.36	3.82	2.46	1.58	0.83	-1.49	-2.52	-3.87	-6.02	-10.46

	2	4.46	3.80	3.22	2.48	0.83	0.04	-1.94	-2.92	-5.09	-6.58	-9.59
	3	4.67	4.29	3.78	2.55	1.40	-0.66	-2.44	-4.95	-6.20	-6.99	-8.86
	4	4.59	4.03	3.46	1.93	1.07	-1.14	-2.52	-4.56	-6.02	-8.86	-20.00
	5	4.49	4.10	3.82	2.12	1.37	0.37	-1.02	-3.87	-5.38	-15.23	-20.00

Table 1: energy propagation when aligned

3.3 Analyzed data

The power in dBm can be converted to mw by the following equation:

$$1\ \text{dBm} = 10 \log_{10} \frac{\text{power}}{\text{mW}}$$

Z (±0.5μm)	trial	0.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
$P_{received}$ (mW)	1	3.00	2.73	2.41	1.76	1.44	1.21	0.71	0.56	0.41	0.25	0.09
	2	2.82	2.67	2.38	1.66	1.36	1.19	0.68	0.48	0.38	0.30	0.01
	3	2.79	2.40	2.10	1.77	1.21	1.01	0.64	0.51	0.31	0.22	0.11
	4	2.93	2.66	2.39	1.80	1.38	0.86	0.57	0.32	0.24	0.20	0.13
	5	2.88	2.53	2.22	1.56	1.28	0.77	0.56	0.35	0.25	0.13	0.01
	6	2.81	2.57	2.41	1.63	1.37	1.09	0.79	0.41	0.29	0.03	0.01
P_{avg} (mW)	/	2.87	2.59	2.32	1.70	1.34	1.02	0.66	0.44	0.31	0.19	0.06

Table 2: energy propagation in mW when aligned

Analysis

Using equation (7) and my experimentally acquired data, I used python 3.11 (code attached in appendix) to plot the theoretical graph for power received with z-axis displacement together with the experimental data to compare:

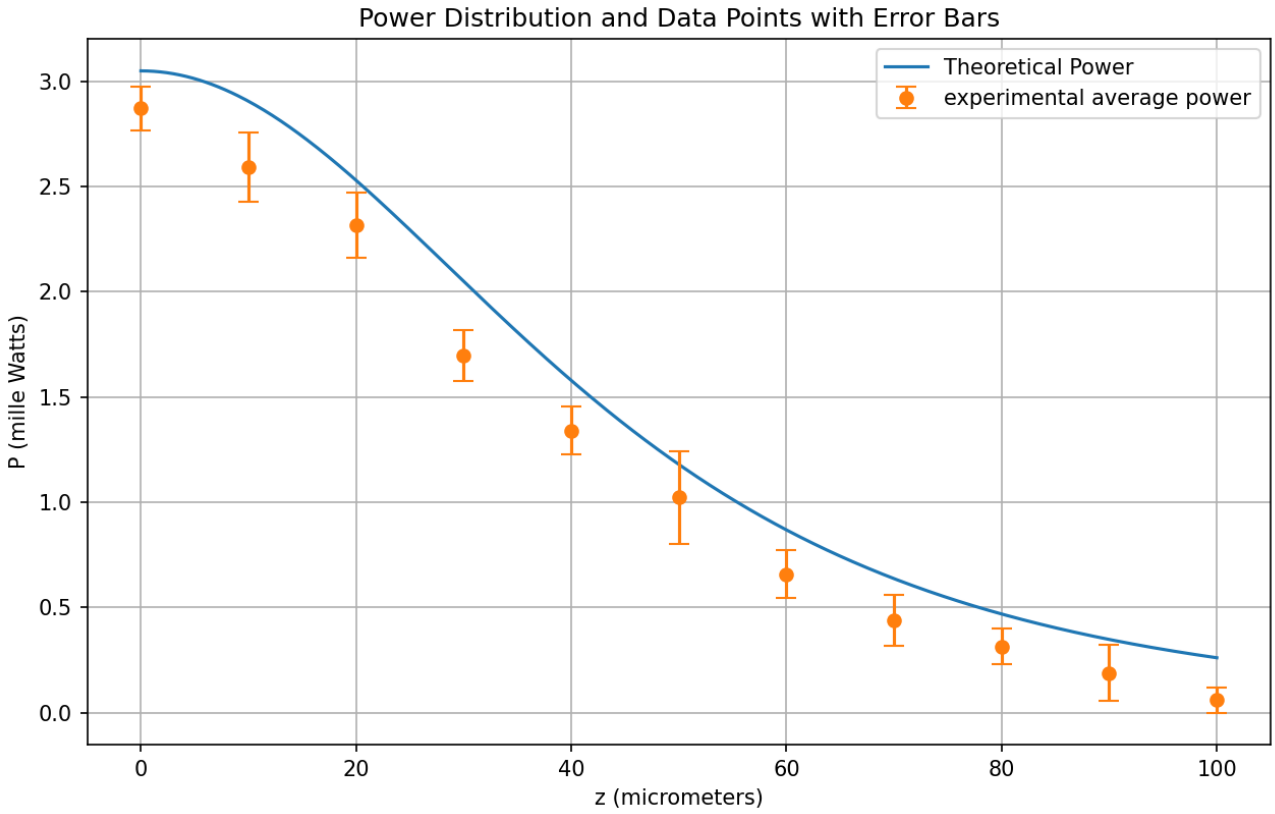


Figure 4: Theoretical and experimental power received vs z-axis displacement
Note: it was too difficult to plot a best fit line for the dots (experimental data) because the pattern is not a regular function.

As expected, the theoretical graph suggests the maximum power received is when the fibers are closest together with the maximum power that could overlap being the power supplied. Also intuitive is the fact that the power received decreases drastically as the fibers are more misaligned, where the power that can be received be negligible when more than 100 micrometer of z-axis misalignments has occurred.

An interesting phenomenon is that the power received changes very little for small misalignment in the 0-3 micrometer range. And the power loss causes by misalignment increases as misalignment increases. Where the maximum slope can be found at 25

micrometers. This means that the effect of z-axis misalignment of less than 5 micrometers is very insignificant on the power received. The power loss caused by z-axis misalignment gets significant after 10 micrometers.

By comparing the experimental data with the theoretical graph two obvious observations occur. The first one is that there seems to be major losses in power regardless of the z value, and the loss is much more than the Fresnel reflection which should only decrease maximum power received by 0.28 mW. During the experiment I already noticed that there seems to be large power loss even when trying to align perfectly. Even though I am doing this at a high school level I made sure that the surface was clean and no external objects were interfering with the experiment. So, a power loss this big might be caused by a combination of misalignment, Fresnel reflection and possibly loss due to beam propagation through air. It provides proof that even during careful experimentation power losses due to alignment are inevitable.

The second observation is that the shape of the data and the theoretical curve correlate very strongly (r = 0.9978.). This provides evidence that signals propagating inside fiber optics is in fact Gaussian and that the Gaussian properties of a beam remain after propagating trough different mediums, since I(r,z) is the beam propagating distribution in air and I(r,0) is the beam propagating distribution in the fiber. Since both were assumed to be Gaussian in order to draw the theoretical graph both assumptions are proven to be true. The reason why a beam keeps its Gaussian properties is the fact that the Gaussian intensity distribution is just a standard distribution with adjusted parameters. Since a light beam consists of an enormous number of light waves it is based on chance as whether a particular wave is at the center of the medium or further from the center. As in a fiber the light is contained by the cladding reflection occurs at the sides of the fiber causing the intensity which is proportional the number of light waves to be concentrated in the middle as the waves have a higher probability of traveling there than the sides where reflection occurs. And as there are enormous

amounts of light waves probability will cause there to be a standard distribution to occur which can be assumed for light beams propagating through a fiber.

Conclusion and Evaluation

By varying the position of two fibers methodically and recording the effect the distance between the two single-mode fibers has on the power that is transferred using a fiber coupling station I have found that: The power loss when aligning two fibers can be modeled using a Gaussian beam and more specifically with the function $\frac{2P_0}{\pi\omega(z)^2}\exp\left(\frac{-2r^2}{\omega(z)^2}\right)$ as only the overlap integral of a beam with a Gaussian intensity distribution will match the experimental power data. This Results in the power loss due to z-axis misalignment being minimal within the 5-micrometer range. When the misalignment is in the beyond the 10-micrometre range the power loss is quite significant. When misalignment is greater than 60 micrometers the power received is minimal.

6.1Weaknesses:

There is a clear unexplained power loss as all my data points are below the theoretical value even though they share exactly the same trend. A partially contributing factor is that during the experiment the source fiber was bent lying on the table/my lap. This is called macro-bending, when a fiber is bent beyond its maximum bend macro-bending power losses will occur which are hard to estimate (Vastag). As the case for my experiment often it was bent beyond its maximum bend however the Gaussian properties of a beam will not be affected as shown by the graph. This will affect the power supplied and thus the P_0 will not be same as the power of the source.

During my experiments I noticed the difference in the amplitude of the fluctuations on the power received on different days. As the fiber coupling station was situated in the

factory a lot of smoke particles might have caused the fluctuations. As on Sundays the fluctuation were noticeably lower than usual due to the relative stillness of the air. In order to be able to determine a reading I always waited 5 seconds per reading to determine the most common reading on the fluctuating power meter. As dBm is converted to mW using a logarithm these fluctuations of 0.1 could cause a large imprecision.

6.2 Strengths:

As I was uncertain about how much the movable holder moved when the dial was turned by 10 units, I needed a way to make sure there was not a large systematic error when moving the fiber. As a fiber is in the micrometer scale, I could not simply measure the displacement using a device. I used a camera with a known enlargement scale. Marking what 10 micrometers would be on the monitor I aligned my fiber with the first mark on the monitor and dialed it until it reached the second mark thus being able to precisely determine how much needed to be dialed for 10 micrometers, which was approximately 10.5 units per 10 micrometers. I was able to enhance and use my creative problem solving during this extended essay writing process.

6.3 Further improvement/investigation:

A main issue encountered when designing the experiment was that the receptor fiber was too large with respect to the beam radius. As result I could only determine what kind of beam it was by using the power received over that area. An alternative method I believe would give me a more precise measurement of the shape and properties of the beam when propagating is using a lens to enlarge the beam. That way the fiber could act as almost point detector allowing me to map out the intensity distribution of planes of z values. With my experiment the fiber could not act as a point detector as the intensity of the beam varied greatly on the surface of the receptor fiber.

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Appendix:

```
from scipy.integrate import quad
import numpy as np
import matplotlib.pyplot as plt

lambda_ = 1310e-9
w0 = 4.6e-6

data = [
    [3, 2.73, 2.41, 1.76, 1.44, 1.21, 0.71, 0.56, 0.41, 0.25, 0.09],
    [2.82, 2.67, 2.38, 1.66, 1.36, 1.19, 0.68, 0.48, 0.38, 0.3, 0.01],
    [2.79, 2.4, 2.1, 1.77, 1.21, 1.01, 0.64, 0.51, 0.31, 0.22, 0.11],
    [2.93, 2.66, 2.39, 1.8, 1.38, 0.86, 0.57, 0.32, 0.24, 0.2, 0.13],
    [2.88, 2.53, 2.22, 1.56, 1.28, 0.77, 0.56, 0.35, 0.25, 0.13, 0.01],
    [2.81, 2.57, 2.41, 1.63, 1.37, 1.09, 0.79, 0.41, 0.29, 0.03, 0.01],
```

```
    [2.871667, 2.593333, 2.318333, 1.696667, 1.34, 1.021667, 0.658333, 0.438333, 0.313333, 0.188333, 0.06]
]

data_np = np.array(data) * 1e13

average = data_np[-1]
errors = np.ptp(data_np[:-1], axis=0)

x_values = np.arange(0, 101, 10)

plt.errorbar(x_values, average, yerr=errors/2, fmt='o', capsize=5, label='Average with Error Bars')

def integrand(r, z):
    term1 = (2 * 0.005) / (np.pi * w0 ** 2 * (1 + (z * lambda_ / (np.pi * w0 ** 2)) ** 2))
    term2 = np.exp((-2 * r ** 2) / (w0 ** 2 * (1 + (z * lambda_ / (np.pi * w0 ** 2)) ** 2)))
    term3 = (2 * 0.005) / (np.pi * w0 ** 2)
    term4 = np.exp((-2 * r ** 2) / w0 ** 2)

    return np.abs(term1 * term2 * term3 * term4 * r) ** 2

z_values = np.linspace(0, 10e-5, 500)  # z from 0 to 10 µm
P_values = []
z_values_micrometers = []

for z in z_values:
    integral, _ = quad(integrand, 0, 4.1e-6, args=(z,))
    P = 2 * np.pi * P0 * integral
    P_values.append(P)
    z_values_micrometers.append(z * 1e6)

plt.figure(figsize=(10, 6))
plt.plot(z_values_micrometers, P_values, label='Theoretical Power')
plt.errorbar(x_values, average, yerr=errors/2, fmt='o', capsize=5, label='experimental average power')

plt.xlabel("z (micrometers)")
plt.ylabel("P (mille Watts)")
plt.title("Power Distribution and Data Points with Error Bars")
plt.gca().yaxis.get_offset_text().set_visible(False)
plt.legend()
plt.grid(True)
plt.show()
```